

SmolVLM

Redefining Small and Efficient Multimodal Models

Hugging Face · Stanford University

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KEY RESULT

0.8 GB

GPU RAM for inference
(batch size = 1)

SmolVLM-256M surpasses
the 300× larger Llafix-80B

Large VLMs Are Powerful but Impractical for Edge Deployment

State-of-the-art VLMs demand massive resources:

- 7B+ parameter models require 27.7 GB GPU RAM
- Cloud dependency introduces latency and privacy risks
- Mobile and edge devices are locked out of VLM capabilities
- Shrinking models naively destroys performance

The gap: No VLM family delivers useful multimodal performance at sub-1GB RAM — until SmolVLM.

Molmo-A1B-7B

7B params

27.7 GB RAM

InternVL2-2B

2B params

~8 GB RAM

SmolVLM-256M

256M params

0.8 GB RAM

SmolVLM-500M

500M params

1.2 GB

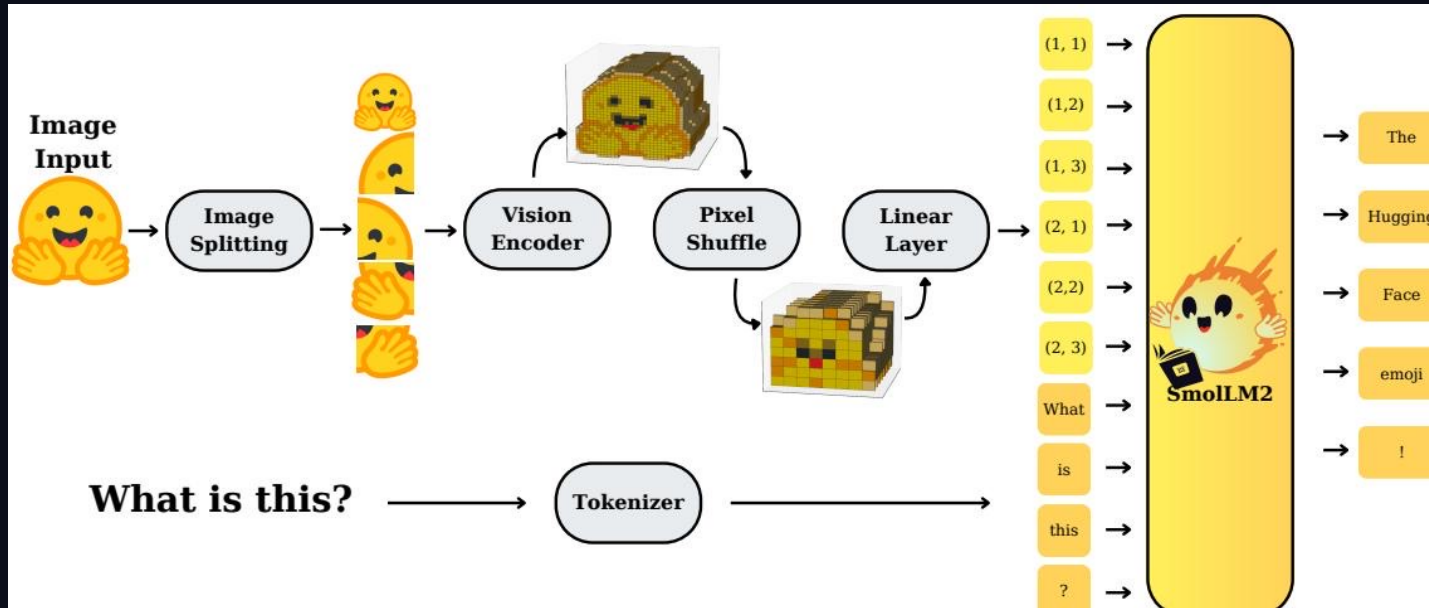
SmolVLM-2.2B

2.2B params

4.9 GB

SmolVLM Architecture: Image Splitting, Pixel Shuffle, and SmolLM2

Backbone



1

Image Splitting

Input images divided into sub-images for processing

2

SigLIP Encoder

Vision encoder extracts spatial features from sub-images

3

Pixel Shuffle

Space-to-depth compression reduces visual tokens by r^2

4

Linear Projection

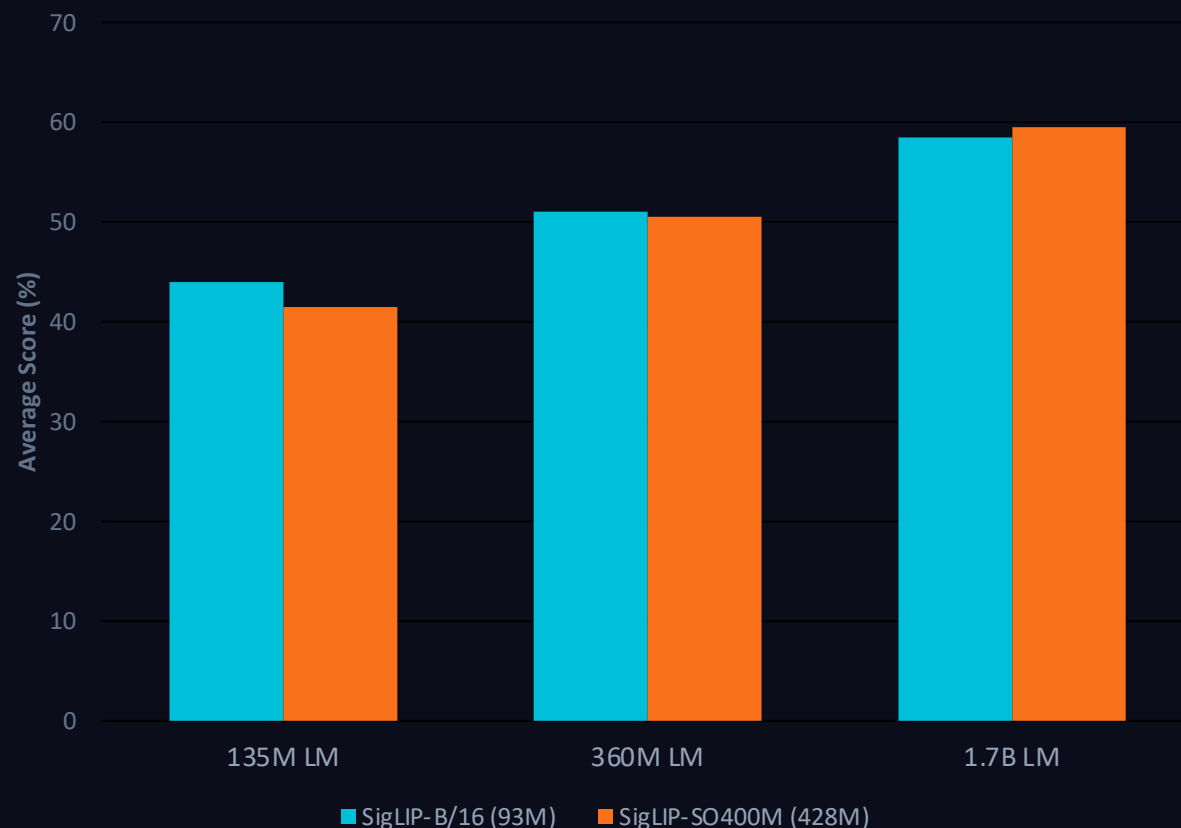
Projects visual features into SmolLM2 embedding space

5

SmolLM2 Backbone

Language model processes concatenated visual + text tokens

Smaller Vision Encoders Outperform Larger Ones in Compact VLMs



KEY INSIGHT

A 93M SigLIP-B/16 paired with a 135M LM **outperforms the 428M SigLIP-SO400M** with the same LM.

Smaller encoders leave more parameter budget for the language model, yielding better VLM performance.

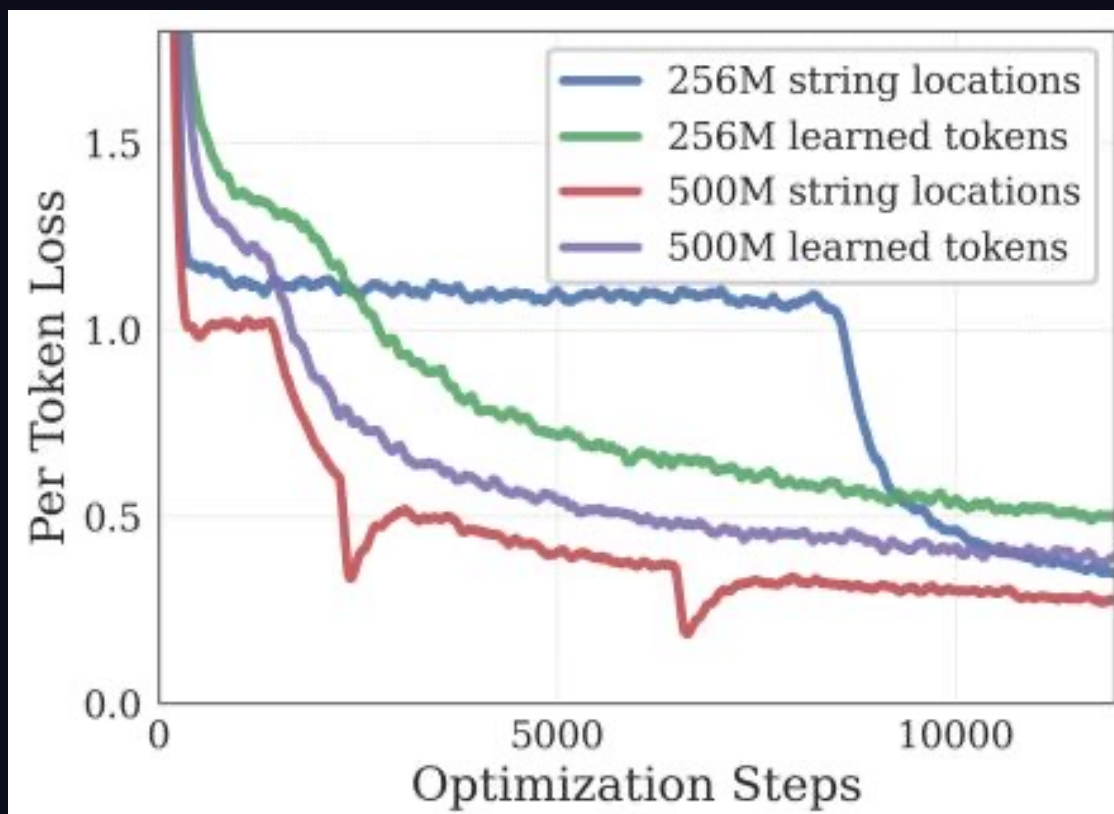
When does the larger encoder win?

Only at 1.7B LM scale does the 428M encoder justify its 10% parameter overhead.

For compact VLMs, smaller encoders are better.

Design Principle: Balance encoder-LM compute allocation

Aggressive Pixel Shuffle (r=4) Cuts Visual Tokens 16× for Small Models



HOW PIXEL SHUFFLE WORKS

Space-to-depth transformation:

- Rearranges spatial pixels into channel dimension
- Reduction factor: r^2 tokens compressed per patch
- **At $r=4$: 16× token reduction**
- At $r=2$: 4× token reduction

Small VLMs benefit from $r=4$ — fewer tokens ease attention overhead and improve long-context modeling.
Larger models prefer $r=2$ for higher fidelity.

$r = 2$

4× reduction

Preferred by larger models
(2.2B)

$r = 4$

16× reduction

Preferred by smaller models
(256M, 500M)

Extending Context from 2K to 16K Tokens Unlocks Higher Image Resolution

1 RoPE Base Adjustment

10K → 273K

Increase Rotary Position Embedding base frequency to support longer sequences without positional confusion.

2 Mixed-Length Fine-Tuning

Long + Short Context

Fine-tune on a blend of long-context and short-context data to maintain performance on both regimes.

3 Per-Size Context

8K / 8K / 16K

SmoIVLM-256M and 500M use 8K context; the 2.2B model extends to 16K for higher-resolution image understanding.

CONTEXT LENGTH IMPACT

Performance improves significantly when extending context from 2K → 16K tokens.

- Longer context allows higher-resolution image encoding — more visual tokens per image
- Enables processing multiple high-res images and video frames in a single forward pass
- The 2.2B model leverages 16K context for best performance; smaller models balance context vs. memory at 8K

SmolVLM Achieves Competitive Performance at Sub-1GB to 4.9GB RAM

Capability	Benchmark	SmolVLM 256M	SmolVLM 500M	SmolVLM 2.2B	Best Efficient OS
Single-Image	OCRBench	52.6%	61.0%	72.9%	54.7% (Molmo-A1B-7B)
Single-Image	AI2D	46.4%	59.2%	70.0%	71.0% (Molmo-A1B-7B)
Single-Image	ChartQA	55.6%	62.8%	68.7%	48.0% (Molmo-A1B-7B)
Single-Image	TextVQA	50.2%	60.2%	73.0%	61.5% (Molmo-A1B-7B)
Single-Image	DocVQA	58.3%	70.5%	80.0%	77.7% (Molmo-A1B-7B)
Single-Image	ScienceQA	73.8%	80.0%	89.6%	87.5% (Molmo-A1B-7B)
Multi-task	MMMU	29.0%	33.7%	42.0%	33.9% (Molmo-A1B-7B)
Multi-task	MathVista	35.9%	40.1%	51.5%	37.6% (Molmo-A1B-7B)
Multi-task	MMStar	34.6%	38.3%	46.0%	43.1% (Molmo-A1B-7B)
Video	Video-MME	33.7%	42.2%	52.1%	45.0% (InternVL2-2B)
Video	MLVU	40.6%	47.3%	55.2%	48.2% (InternVL2-2B)
	Average	44.0%	51.0%	59.8%	—

RAM USAGE

0.8–4.9 GB vs. 27.7 GB (Molmo-A1B-7B)

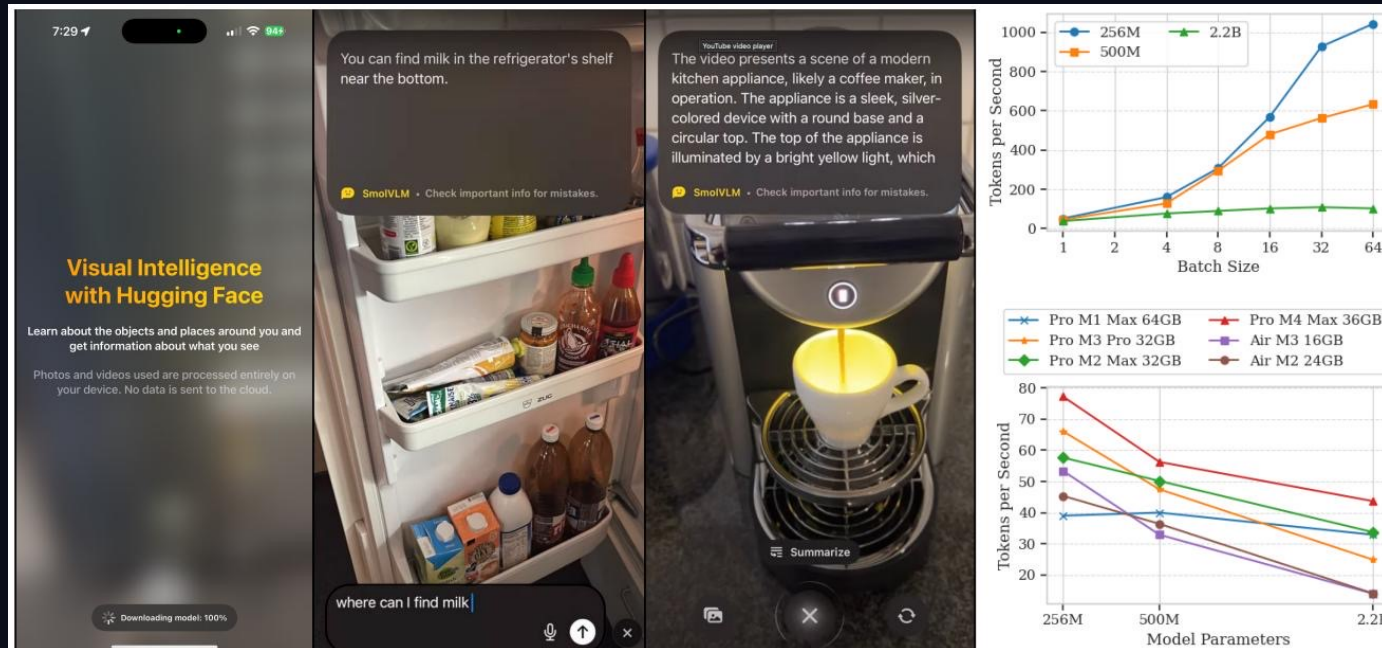
SCALE EFFICIENCY

256M surpasses the 300× larger Idefics-80B

AVERAGE SCORES

256M: 44.0% · 500M: 51.0% · 2.2B: 59.8%

Real-Time Inference on iPhones: Up to 80 Tokens/Second on Apple Silicon



ON-DEVICE THROUGHPUT

~80 t/s

SmolVLM-256M on Pro M1 Max 64GB (batch=1)

~55 t/s

SmolVLM-256M on MacBook Air M3 16GB (batch=1)

~1,000 t/s

SmolVLM-256M on high-end devices (batch=64)

HuggingSnap iOS App

Processes photos & videos entirely locally — no cloud dependency

Key Takeaways: Design Principles for Efficient Multimodal Models

0 Balance Encoder-LM Compute

1
Smaller vision encoders (93M) outperform larger ones (428M) for compact VLMs — leave parameter budget for the LM.

0 Extend Context Length

2
Scaling from 2K → 16K tokens via RoPE adjustment and mixed-length fine-tuning unlocks higher image resolution.

0 Use Aggressive Token Compression

3
Pixel shuffle with $r=4$ delivers 16× token reduction for small models, easing attention overhead and improving long-context modeling.

0 Sub-1GB Inference Is Achievable

4
SmolVLM-256M runs at 0.8 GB GPU RAM, surpassing the 300× larger Llafix-80B — architectural efficiency beats raw scale.

0 Fully Open-Source

5
All model weights, training data, code, and the HuggingSnap mobile app are released to promote reproducibility and on-device AI research.